ECE 329 **Homework 9** **Due: Mon, Dec 1, 2025, 11:59PM**

4 problems totaling 30pts with 10pts extra Bonus Points. (Bonus problems are optional and the earned bonus credits will be used to "fill in" the "missing points" in the total homework score.)

1. (10pts) **Circular polarized wave problem:**

   a) For phasor $\tilde{\mathbf{E}} = (2\hat{x} + j2\hat{y})e^{-jkz}$:

      i. (1.5pts) Does its $\hat{x}$ component *lead* or *lag* its $\hat{y}$ component by 90°? You can determine the answer by inspecting the locations of complex numbers $E_x \equiv \tilde{\mathbf{E}} \cdot \hat{x}$ and $E_y \equiv \tilde{\mathbf{E}} \cdot \hat{y}$ on the complex plane. The $\hat{x}$ component E_x *leads* the $\hat{y}$ component E_y by 90° if the angle of (complex number) E_x (at, say, $z = 0$) is greater than the angle of E_y by 90° (otherwise, it *lags*).

      ii. (1.5pts) Does the time-harmonic expression corresponding to phasor $\tilde{\mathbf{E}} = (2\hat{x} + j2\hat{y})e^{-jkz}$ describe a right-hand-circular (RHC) or left-hand-circular (LHC) wave?

      Hint: once the rotation direction of the vector is figured out, RHC or LHC can be deduced by identifying the wave propagation direction and pointing your right thumb in the propagation direction — the wave is then RHC if your right hand fingers curl in the direction of the rotation of the vector or else LHC if your fingers curl against the rotation direction.

   Try understanding that in circularly polarized waves, the field vector rotates from that coordinate axis having the "leading phasor component" to the other axis having the "lagging phasor component" (by 90 degrees) just as illustrated by parts (a) and (b) above.

   b) (3pts) Is the magnetic field intensity phasor $\tilde{\mathbf{H}} = (\hat{z} + j\hat{y})e^{+jkx}$ describing a RHC or LHC TEM wave?

   Hint: apply what you learned in part (a) to $\tilde{\mathbf{H}}$ — in circularly polarized waves **E** and **H** field vectors will always rotate in the same direction!

   c) (4pts) Determine the wave polarization of the waves/phasors given on Problem 1 of HW 8, using the following categories: linear polarized (state the polarization direction), circular polarized (state RHC or LHC), or neither (elliptical polarized).

2. **(Bonus Problem: 10pts) Wave polarizations:**

Two current sheets are fixed in free-space in the $x = 0$ and $x = \lambda/4$ planes having surface current densities $\mathbf{J}_{s1}(t) = J_1 \cos(\omega t + \phi)\hat{z}$ and $\mathbf{J}_{s2}(t) = J_2 \cos(\omega t)\hat{y}$, respectively, where J_1, J_2, and ϕ are real numbers. TEM waves will propagate away from each sheet and superpose with each other to create a wavefield whose polarization depends on the value of ϕ and the ratio of J_2 and J_1.

a) (2pts) Determine the phasors corresponding to the surface current densities $\mathbf{J}_{s1}(t)$ and $\mathbf{J}_{s2}(t)$, respectively.

b) (2pts) Determine the phasors of the electric fields radated from each of the current sheet in the region $x > \lambda/4$.

Assuming that $J_2 = J_1$, determine the value(s) of ϕ for the superposed TEM waves propagating in the region $x > \lambda/4$ to have:

a) (2pts) right handed circular polarization,

b) (2pts) left handed circular polarization,

c) (2pts) linear polarization.

Justify your choice of ϕ in each case by determining the electric field phasor produced for the region $x > \lambda/4$ with your choice of ϕ.

3. (10pts) **Plane wave reflections:** A plane wave field

$$\mathbf{E} = 4\sin(\omega t - \beta z)\hat{y}\,\frac{\mathrm{V}}{\mathrm{m}}$$

is propagating in vacuum in $+z$ direction and is incident on $z = 0$ plane which happens to be the boundary of a perfect dielectric having permittivity $2.25\epsilon_o$ and permeability μ_o in the region $z > 0$. Calculate:

a) (3pts) The reflected and transmitted electric fields in terms of reflection and transmission coefficients $\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ and $\tau = 1 + \Gamma$ for the described interface. Express the fields in phasor form.

b) (3pts) The incident, reflected, and transmitted **H** fields in A/m units. Express the fields in phasor form.

c) (4pts) The time-average power per unit area transported by the incident, reflected, and transmitted waves in $\mathrm{W/m^2}$ units. Are the calculations compatible with energy conservation principle? Explain.

4. (10pts) **Telegrapher's equations**

$$\begin{aligned} -\frac{\partial V}{\partial z} &= \mathcal{L}\frac{\partial I}{\partial t} \\ -\frac{\partial I}{\partial z} &= \mathcal{C}\frac{\partial V}{\partial t} \end{aligned}$$

govern the voltage and current waves $V(z,t)$ and $I(z,t)$ that propagate on transmission line systems.

If $V(z,t) = 6\cos(\omega t + \beta z)$ on a TL, determine $I(z,t)$ and β (a positive number) by using the telegrapher's equations twice.

Hint: First use one of the telegrapher's equations to determine $I(z,t)$. Then use the other telegrapher's equation to determine $V(z,t)$ from $I(z,t)$ found in the first step. By requiring $V(z,t)$ found in step 2 to equal the original $V(z,t)$ you should be able to identify β.